ID	Requirement	Description	Story Points	Priority	Sprint Number
R1	Platform & Architecture Research	Research and validate optimal architecture using Android Studio with Unity SDK bridging to Meta Quest Pro.	13	P0	1
R2	SDK & Toolchain Validation	Evaluate Meta Quest Pro SDKs, Unity XR pipeline, Android build targets, and integration constraints.	8	P0	1
R3	Data Flow Design	Define end-to-end data flow between phone sensors, Android app, Unity bridge, and headset.	5	P0	1
R4	Security & Privacy Baseline	Research Android health data access, permissions, encryption options, and local-only storage models.	5	P0	1
R5	HUD Overlay Baseline (MVP)	Implement a minimal in-headset HUD overlay with proper placement, anchoring, and visibility controls using Unity.	13	P0	2
R6	Widget System Core	Implement modular widget framework supporting widget creation, placement, repositioning, and removal.	8	P0	2
R7	HUD Performance Optimization	Optimize rendering and update loops to maintain stable ≥72 FPS on Meta Quest Pro.	8	P0	2
R8	Session Persistence	Persist HUD layout, widget positions, and configuration across sessions.	5	P1	2
R9	Widget SDK Contract Definition	Define formal widget lifecycle interface, size constraints, performance budgets, and isolation rules to enable parallel module development.	8	P1	3
R10	Dashboard Layout Manager Enhancement	Improve slot layout system for scalable positioning, resizing, and world-space alignment.	8	P1	3
R11	Global Event Bus Implementation	Implement publish/subscribe messaging system for widget communication without direct coupling.	8	P1	3
R12	Android App Skeleton	Create Android Studio project with modular architecture, module layout, and build configuration.	5	P1	3
R13	Permission & Consent Framework	Implement explicit, revocable permission handling for health and sensor data on Android.	5	P1	3
R14	Data Serialization Layer	Define structured, versioned data models for metrics to be sent to the headset.	5	P1	3
R15	Local Data Handling	Ensure all collected data remains local by default with clear lifecycle and retention rules.	5	P1	3
R16	Connection Reliability	Handle connection loss, retries, and safe fallback states between Android and Unity.	13	P2	4
R17	Latency & Smoothing	Implement smoothing and buffering to keep perceived data latency under 3 seconds.	8	P2	4
R18	Telemetry & Debug Logging	Add internal logging for data flow timing, dropouts, and message integrity.	8	P2	4
R19	Dashboard Animation & Resize System	Implement smooth widget expand/minimize transitions with state persistence.	13	P2	4
R20	Health Widget Module	Develop independent heart rate widget using SDK contract with update states and failure handling.	13	P2	4
R21	Extended Health Widget	Integrate additional health metrics such as steps, calories, HRV, or SpO_2 within modular widget framework.	8	P2	5
R22	GPS Widget Module	Develop phone-sourced GPS speed and heading widget with graceful failure handling.	8	P2	5
R23	Safety Mode HUD	Add low-distraction HUD mode with optional dimming and reduced motion effects at dashboard level.	8	P3	5
R24	Error Isolation Framework	Ensure individual widget failure does not crash or freeze dashboard system.	8	P3	5
R25	Cross-Device Compatibility	Test functionality across multiple Android devices and sensor sources where possible.	5	P3	5

ID	Requirement	Description	Story Points	Priority	Sprint Number
R26	Line-of-Sight Mapping (Experimental)	Investigate gaze-based association of nearby users with strict fail-safe behavior.	13	P3	6
R27	Proximity Discovery (Experimental)	Explore opt-in nearby presence detection using anonymous identifiers.	8	P3	6
R28	Shared Stats Rendering	Render minimal shared stats with explicit, mutual user consent.	8	P3	6
R29	System Refactor & Cleanup	Refactor dashboard core and modular widget architecture for stability and maintainability.	8	P3	6
R30	Integration Hardening	Stabilize end-to-end system integration across dashboard, widgets, permissions, and device communication.	8	P4	6
R31	Proximity Security Controls	Enforce mutual consent, session-based opt-in, and immediate revoke controls.	5	P4	6
R32	Ethical & Privacy Evaluation	Evaluate privacy, misuse risk, and ethical implications of social AR features.	5	P4	6
R33	NSDK 4.0 Integration & Validation	Work toward making Niantic Spatial NSDK 4.0 functional within the project pipeline and validate compatibility with the existing XR stack.	13	P4	6
R34	Deployment Readiness Cleanup	Clean up project structure, remove deprecated assets and scripts, and prepare the codebase for reliable deployment.	8	P4	6
R35	Build Packaging & Environment Setup	Finalize build settings, dependency management, and environment configuration to ensure consistent deployment across systems.	8	P4	6
R36	Release Testing, Documentation & Demo Prep	Complete final testing, deployment validation, documentation, and presentation readiness for handoff and demonstration.	8	P4	6